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Comparing Machine-Learning Models for Enterprise Bankruptcy Prediction

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Abstract

This thesis develops and assesses a machine-learning framework for one-year-ahead corporate bankruptcy prediction in Italy using AIDA financial-statement data for 2016–2022. The study proceeds in two phases to address an extreme class imbalance where failures represent only 0.127% of firm-year observations.

Phase 1 compares six algorithms on a balanced pilot sample (1:1 ratio of bankrupt to healthy firms). Random Forest, XGBoost, and LightGBM deliver ROC-AUC values above 0.93. Hyperparameters for every model are tuned with Bayesian optimisation (Optuna TPE).

Phase 2 extends the two most efficient models (XGBoost and LightGBM) to the full population of roughly 190 000 enterprises. The Optuna search maximised ROC-AUC while treating the retention ratio of healthy firms as a tunable hyperparameter, exploring down-sampling levels from 0.5% to 100%. The best configuration retained the entire non-bankrupt population and applied a minority-class weight of 778 in the loss function, with all hyperparameters optimized through 700 Bayesian trials.

This ROC-centred approach relates to ongoing debates in the imbalanced classification literature. Earlier work recommended PR-AUC for imbalanced datasets, but recent studies question this guidance for extreme imbalance cases. Our experiments indicate that ROC-AUC optimization produced more stable test performance than alternative metrics when the failure rate is 0.127%.

In Phase 2, the tuned XGBoost model achieved a test ROC-AUC of 0.947 and, at 90% recall, flags only 11.2% of firms for review, an eight-fold reduction in investigative effort relative to random selection. Among firms predicted as healthy, the false negative rate drops approximately to 0.014%, one-ninth of the base 0.127% failure rate.

While the framework offers actionable insights for lenders and regulators, it is limited to pre-2023 data and financial ratios; future work could incorporate judicial-liquidation outcomes introduced by recent legal reforms.

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Chapter 1

Introduction and Literature Review

1.1 Motivation and Research Questions

Corporate bankruptcy imposes sizeable private and social costs, including job losses, write-offs in the banking sector, and the deterioration of local supply-chain networks. The timely *ex-ante* identification of vulnerable companies is therefore of significant interest to a wide range of economic actors:

- **Financial Institutions:** For lenders and banks, an effective early-warning system is fundamental to credit risk assessment, enabling more accurate pricing of loans and the prevention of significant capital losses.
- **Investors and Shareholders:** For investors, the ability to predict bankruptcy is crucial for portfolio management, guiding decisions to divest from at-risk assets or to identify fundamentally sound companies.
- **Supply Chain Partners:** For non-financial enterprises, the unexpected failure of a key supplier or customer can trigger severe operational disruptions. Predictive models offer a tool for managing supply chain risk and ensuring business continuity.
- **Policymakers and Regulators:** For governmental bodies, monitoring the aggregate health of the corporate sector is essential for economic stability. These models can help identify systemic risks and inform potential policy interventions.

This thesis develops and evaluates a two-phase machine learning framework to tackle this task within the specific context of the Italian economy, confronting the core challenge of extreme class imbalance where bankruptcies represent only 0.127% of firm-year observations. To that end, this research will address the following key questions:

1. **The Performance Question:** Using financial statement data, how accurately can machine learning models predict corporate bankruptcy for Italian firms one year in advance under realistic conditions of extreme class imbalance? What is the practical business utility of such a model in reducing investigative workload?
2. **The Algorithm Selection Question:** In a preliminary analysis on a balanced sample, which algorithmic approaches—from linear models to tree-based ensembles and neural networks—provide the best predictive performance, identifying suitable candidates for a large-scale, computationally intensive analysis?
3. **The Methodological Question:** What is a robust and scalable framework for a one-year-ahead bankruptcy prediction task on a full, imbalanced population? This involves two critical sub-questions:
 - (a) What is the optimal strategy for handling an extreme minority class (0.127%), comparing various sampling and weighting techniques?
 - (b) Which optimization metric (e.g., ROC-AUC, PR-AUC) yields the most stable and reliable model performance, especially given recent debates in the machine learning literature on imbalanced classification?

Answering these questions will provide a comprehensive evaluation of a complete modeling pipeline, from initial algorithm selection to final deployment on a real-world dataset defined by its scale and extreme class imbalance.

1.2 The Evolution of Bankruptcy Prediction Models

The academic pursuit of bankruptcy prediction began with the fundamental premise that financial distress is not a random event, but a process with observable antecedents. The earliest research focused on identifying these signals within standard financial statements. The pioneering work of Beaver (1966) used *univariate analysis* to demonstrate that individual financial ratios differed systematically between failed and non-failed firms, establishing an empirical basis for all subsequent research.

The field took a significant leap forward with the application of multivariate statistical techniques. The most influential of these early models is the Z-Score developed by Altman (1968), which applied *Multivariate Discriminant Analysis (MDA)* to combine several financial ratios into a single, actionable score to classify firms. A further crucial refinement came from Ohlson (1980), who introduced *logistic regression (logit)*. This shifted the objective from simple classification to a more flexible *probabilistic* framework, which relaxed the strict statistical assumptions of MDA (such as multivariate normality) that are often violated by financial data.

These foundational models represent what Breiman (2001) famously termed the "data modeling culture", where the primary goal is to find an interpretable, underlying stochastic model that explains the data. However, for a phenomenon as complex as corporate bankruptcy, Breiman argued for an "algorithmic modeling culture", which prioritizes predictive accuracy above all else, treating the true data-generating process as an unknowable "black box."

This modern paradigm, enabled by vast increases in data availability and computational power, has led to the widespread adoption of machine learning algorithms. Systematic reviews, such as that by Alaka et al. (2018), confirm that techniques like Support Vector Machines, Random Forests, and Gradient Boosting Machines now dominate the field, consistently demonstrating superior predictive performance (Barboza et al., 2017). This trend holds within the Italian context, where studies such as Bragoli et al. (2021) have explored the utility of machine learning models and the incremental value of industrial variables.

While the superiority of algorithmic models is well-established, their application to real-world bankruptcy datasets introduces significant methodological challenges, most notably the problem of extreme class imbalance. For many years, the consensus in the machine learning literature recommended the Area Under the Precision-Recall Curve (PR-AUC) as a more informative metric than the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) in such settings. However, recent research has begun to question this guidance, particularly in cases of extreme rarity. Studies by Holzmann and Klar (2024), Richardson et al. (2024), and McDermott et al. (2024) argue that ROC-AUC can provide a more stable and reliable assessment of model performance, as PR-AUC can become unstable and less discriminative when the positive class is exceptionally small. This thesis engages with this debate by providing an empirical analysis of which evaluation framework yields more robust results for the specific, highly imbalanced dataset used in this study.

1.3 Methodological Context and Thesis Outline

Building on the established literature, the framework developed in this thesis is explicitly designed to address the specific and complex challenges inherent in predicting corporate bankruptcy with real-world Italian data. The methodology was shaped by three core considerations: defining a relevant and manageable study population, navigating the practical and statistical issues of rare events, and ensuring the final model is a useful tool for economic decision-making.

First, the study population was focused on Italian enterprises that reported more than 14 employees in at least one year of the observation period. This choice was driven by two factors. Primarily, the 15-employee threshold represents a significant dividing line in Italian labor law, which imposes different regulatory and cost structures on firms above this size. This creates a more homogeneous group for analysis. Secondly, this focus allowed for a manageable data

extraction process from the comprehensive AIDA database.

Second, the project confronted the challenge of extreme class imbalance through a two-phase research design. **Phase 1** addresses the algorithm selection question by comparing six distinct machine learning models on a balanced pilot sample. This controlled environment allowed for an initial performance benchmark and the identification of the most efficient and powerful algorithms. **Phase 2** then applies the top-performing models to the core analytical challenge: the full population of nearly 190 000 firms, where the bankruptcy rate is a mere 0.127%. This phase involved extensive experimentation with sampling and weighting strategies to handle the imbalance and a rigorous hyperparameter optimization to maximize predictive power on the complete, realistic dataset.

Finally, while acknowledging the debate around evaluation metrics, the primary technical objective of this thesis is to produce the best possible Receiver Operating Characteristic (ROC) curve. In the context of rare-event data where standard metrics like accuracy are misleading, the ROC curve is a particularly robust evaluation tool. The Area Under the Curve (AUC) measures a model's pure discriminatory power in a way that is insensitive to the class imbalance in the evaluation set, as its axes (True Positive Rate and False Positive Rate) are rates calculated independently within each class (Fawcett, 2006). This theoretical stability makes it a strong candidate for optimization, a choice that is empirically tested in this thesis.

Furthermore, the full ROC curve is essential for problems with asymmetric misclassification costs. In this context, the cost of a False Negative (a missed bankruptcy) is substantially higher than the cost of a False Positive (a healthy firm flagged for review). The ROC curve allows a decision-maker to visualize the trade-off between sensitivity and specificity and select an operating threshold that aligns with their specific risk tolerance and cost structure, rather than relying on a statistically arbitrary one.

To present this work in a structured manner, the remainder of this thesis is organized as follows. **Chapter 2** details the data and the complete two-phase methodological framework. It describes the AIDA data source, preprocessing, feature engineering, and the construction of the target variable. It then outlines the experimental design for both Phase 1 (balanced sample comparison) and Phase 2 (full population analysis), including the sampling experiments, the metric selection process, and the hyperparameter optimization strategy. **Chapter 3** presents the empirical results. It begins with the findings from Phase 1, leading to the selection of XGBoost and LightGBM. It then presents the main results from the Phase 2 analysis, including the final model's performance on the full test set, an analysis of the metric stability experiment, a discussion of the model's business value, and an examination of key feature importances. Finally, **Chapter 4** concludes by summarizing the findings, acknowledging limitations, and proposing future research directions, such as the incorporation of recent *Liquidazione Giudiziale* (judicial liquidation) events.

Chapter 2

Data, Methodology, and Experimental Design

2.1 Data Source, Feature Engineering, and Target Definition

The empirical analysis is based on a comprehensive dataset constructed from the AIDA (Analisi Informatizzata Delle Aziende Italiane) database, provided by Moody's. The full data extraction was a significant undertaking, requiring two weeks of scripted processing. Given this, and the high computational cost associated with training models like Support Vector Machines and Neural Networks, a two-phase research design was adopted. A smaller, balanced pilot study was first conducted to validate the data's potential and the general viability of the modeling approach before proceeding with the computationally intensive analysis of the entire population. This data processing pipeline was implemented using the Python programming language, primarily relying on the Pandas and Numpy libraries for data manipulation.

The raw data covered the financial years 2015–2024 for a population of approximately 190 000 Italian enterprises. This population was defined to include any enterprise reporting more than 14 employees in at least one year, a threshold chosen to focus the analysis on firms operating under a more homogeneous set of Italian labor regulations. The initial extraction was in a **wide format**, containing 496 distinct variables for each of the 10 years, resulting in nearly 5 000 columns (e.g., `REVENUE_2016`, `REVENUE_2017`, etc.). This structure was transformed into a **panel (long) format**, where each row represents a unique firm-year observation, yielding approximately 1.6 million observations before cleaning.

2.1.1 Feature Engineering and Selection

After reshaping the data into a panel structure, variables with more than 95% missing values across all observations were removed. As a pragmatic feature reduction step, this serves a dual purpose: it prevents the introduction of excessive noise through imputation, which can harm model performance, while also significantly decreasing the computational time and complexity of the subsequent modeling steps (Kuhn and Johnson, 2013). This step reduced the initial set of 496 variables to 394 core features. Further feature engineering was performed to enrich the dataset:

- **Company Age:** Calculated from the firm’s incorporation year.
- **Legal Form:** The categorical `Legal form` variable was converted into 32 binary dummy variables.
- **Seasonality:** Date-based features, such as the balance sheet closing date, were converted to month-only to capture potential seasonal patterns without introducing year-specific information.
- **Geography:** The latitude and longitude of the firm’s registered office were retained. These were the only time-invariant (static) variables intentionally kept in the feature set, allowing the models to capture potential location-based economic patterns.

The extensive set of financial and business ratios used in the analysis were extracted directly from the AIDA database. It is important to note that these ratios were included in the full Phase2 data extraction, not the initial pilot sample. After the addition of the 32 dummy variables, the final feature set used for modeling contained **426 variables**.

2.1.2 Construction of the Dependent Variable

The objective of this thesis is to predict bankruptcy one year in advance. The dependent variable was constructed using the "events" data within AIDA. For the scope of this study, a firm was labeled as a failure (`Target = 1`) exclusively based on the occurrence of a **‘Fallimento’** (Bankruptcy) event. The newer `Liquidazione Giudiziale` (Judicial Liquidation) status, introduced by recent legal reforms, was noted but deliberately excluded from the target definition to maintain a consistent failure definition and is reserved for future analysis.

To prevent data leakage, a strict temporal logic was applied. If a firm went bankrupt in year t , its observation for year $t - 1$ was assigned `Target = 1`. Subsequently, all observations for that firm from year t onwards (i.e., the year of bankruptcy and any following years) were completely removed from the dataset.

2.1.3 Final Dataset and Temporal Splitting

The final dataset for the main analysis was constructed using feature data from the years 2016 to 2021. Data from 2015 was excluded due to inconsistencies in the source "events" variable. Event data from 2022 was used solely to label the 2021 observations, after which the 2022 feature data was discarded. Feature data from 2023–2024 was not used as these years predominantly contained ‘Liquidazione Giudiziale’ events.

The dataset was split chronologically to ensure that the model is always validated and tested on data that is "in the future" relative to its training data. This cross-temporal validation approach provides a more realistic estimate of a model’s real-world performance. The split, shown in Figure 2.1, is as follows:

- **Training Set:** 2016–2018
- **Validation Set:** 2019
- **Test Set:** 2020–2021

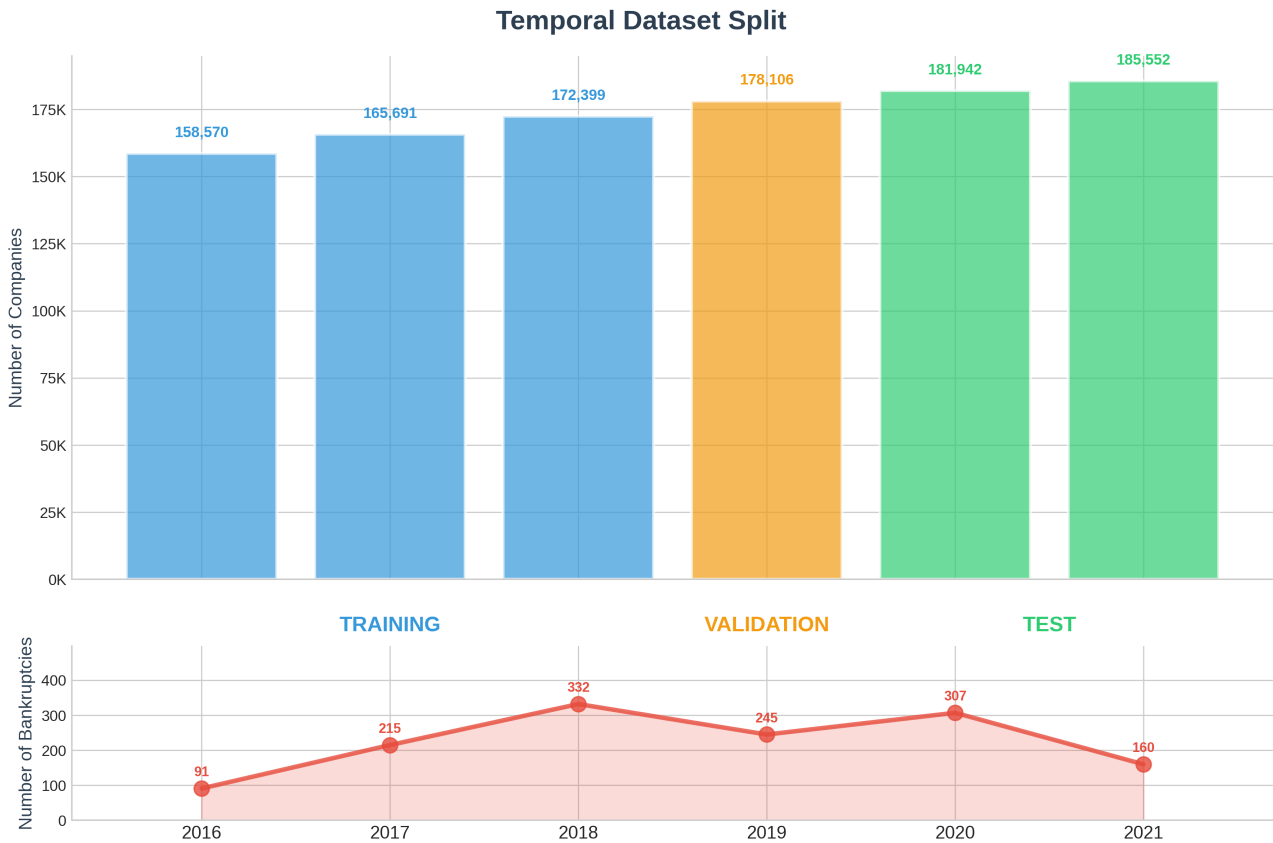


Figure 2.1: Temporal split of the dataset into training, validation, and test sets. The top panel shows the number of active companies per year, while the bottom panel shows the corresponding number of ‘Fallimento’ (Bankruptcy) events.

2.2 Phase 1: Algorithm Selection on a Balanced Sample

The first phase of this research was designed as a pilot study to efficiently compare a wide range of machine learning algorithms. The goal was to identify the most promising candidates for the computationally intensive analysis on the full, imbalanced dataset in Phase 2. This required creating a smaller, more manageable dataset with a balanced class distribution.

2.2.1 Balanced Sample Construction

To construct the pilot dataset, all **1,350** firm-year observations flagged as ‘Fallimento’ (the bankrupt class) were selected. An equal number of healthy (non-bankrupt) firms were then added, chosen via random sampling from the AIDA database. This process yielded a balanced dataset of **2,700** unique enterprises with a 1:1 class ratio, totaling approximately 13 000 firm-year observations after including their available historical data. This balanced sample was partitioned using the following temporal split, a structure maintained for consistency throughout the study:

- **Training Set:** 2016–2018
- **Validation Set:** 2019
- **Test Set:** 2020–2021

2.2.2 Preprocessing Pipeline for Model Comparison

A critical requirement for this phase was to ensure a fair and controlled comparison across all six selected algorithms. Models such as **Logistic Regression**, **Support Vector Machines (SVM)**, the **Multi-layer Perceptron (MLP)**, and the standard implementation of **Random Forest** in scikit-learn cannot natively handle missing data. Therefore, a universal preprocessing pipeline was developed and applied to all models in Phase 1.

Hierarchical Imputation

To address missing values without introducing lookahead bias or data leakage from future periods, a three-level hierarchical imputation strategy was implemented. For a missing value in year t for firm i :

1. **Linear Trend Extrapolation:** The primary method was to estimate the value by extrapolating a linear trend from that firm’s data in years $t - 1$ and $t - 2$.
2. **Last Observation Carried Forward (LOCF):** If a two-year history was unavailable, the missing value was filled with the last known value from the preceding year.

3. **Global Median Fallback:** Only if a firm had no prior data for a feature was the value imputed using the global median. This median was calculated *exclusively* from the training set.

Feature Scaling

For algorithms sensitive to the scale of input features (**Logistic Regression, SVM, MLP**), all numeric features were standardised to have a zero mean and unit variance. The scaler was fitted only on the training data.

2.2.3 Algorithm Specification and Optimization Framework

Six distinct models were selected to represent a diverse set of learning strategies, implemented using established open-source libraries:

- **Logistic Regression:** A linear model with an L1 (Lasso) penalty for regularization, implemented using **Scikit-learn**.
- **Random Forest:** An ensemble of decision trees based on bagging, also provided by the **Scikit-learn** library.
- **XGBoost & LightGBM:** Two state-of-the-art gradient boosting implementations, used via their dedicated high-performance Python libraries.
- **Support Vector Machine (SVM):** A margin-based classifier, testing both linear and non-linear (RBF) kernels, implemented in **Scikit-learn**.
- **Neural Network:** A Multi-Layer Perceptron (MLP) deep learning architecture, built using the **TensorFlow** library.

Even with a 1:1 ratio of bankrupt to healthy firms, the number of firm-year observations for the healthy class was much larger. To handle this remaining imbalance in the training set, cost-sensitive learning was applied at the algorithm level. Each model was balanced using its specific parameter: `class_weight='balanced'` was set for the scikit-learn models (Logistic Regression, RandomForest, and SVM); a manually calculated weight dictionary was passed to the Neural Network's loss function; `scale_pos_weight` was used for XGBoost; and `is_unbalance=True` was set for LightGBM. For each model, the optimization objective was to maximize the **ROC-AUC** score on the 2019 validation set. Because the ROC-AUC metric is insensitive to class imbalance, it provides a reliable measure of pure discriminatory power, making it an ideal metric for comparing different algorithmic architectures in this balanced pilot phase.

2.2.4 Transition to Phase 2

This entire preprocessing pipeline—imputation and scaling—was exclusive to Phase 1. As Phase 2 focused solely on **XGBoost** and **LightGBM**, the pipeline was abandoned. These gradient boosting libraries have a robust internal mechanism for handling missing values, learning a default direction for missing observations during tree splits. This allows the model to treat the absence of data as potentially useful information, making an external imputation step unnecessary.

2.3 Phase 2: Full Population Analysis under Extreme Imbalance

After identifying XGBoost and LightGBM as the most promising candidates, Phase 2 addressed the core research challenge: building a predictive model on the full, unfiltered population of Italian enterprises. This phase confronted the problem of extreme class imbalance directly.

2.3.1 Full Population Dataset and the Imbalance Challenge

The analysis in this phase used the entire dataset, partitioned according to the temporal split established in Phase 1. As shown in Figure 2.2, the class imbalance is severe. The training set (2016–2018) contains **496 660** firm-year observations, of which only **638** are bankruptcies. This corresponds to a failure rate of just **0.128%**, or approximately 1 in every 778 observations. This extreme rarity poses a significant challenge for machine learning algorithms, which can easily achieve high accuracy by simply predicting the majority class (‘healthy’) for all cases.

The validation and test sets were, crucially, the full, unfiltered populations for their respective years, ensuring that the final model evaluation reflects a true, real-world deployment scenario.



Figure 2.2: Dataset composition for the training, validation, and test sets. The extreme rarity of the bankrupt class (red) is evident across all partitions.

2.3.2 Experimental Design for Imbalance Handling

The extreme class imbalance demanded a careful handling strategy. While methods for generating synthetic data, such as the Synthetic Minority Over-sampling TEchnique (SMOTE), are common, they were deliberately avoided. These techniques would have required a full imputation of the dataset beforehand, which would destroy any potential information contained within the patterns of missing data. Furthermore, validating the business sense of synthetically generated data points across 426 financial variables without deep domain expertise would be highly problematic.

Instead, a sampling optimization strategy was employed. Rather than pre-selecting a fixed sampling ratio for the healthy class, the **down-sampling ratio itself was treated as a hyperparameter to be tuned by Optuna**. This allowed the optimization process to simultaneously find the best model parameters and the best data sampling configuration.

The experiment was set up as follows:

1. A range of down-sampling ratios for the healthy class was defined: [5%, 10%, 30%, 50%, 70%, 90%, 100%].
2. During each Optuna trial, the optimizer selected one of these ratios.

3. A training set was constructed using all 638 bankrupt firms and the chosen percentage of healthy firms.
4. For this newly constructed sample, a corresponding ‘scale_pos_weight’ was calculated as the ratio of healthy to bankrupt observations and passed to the model.
5. The model was then trained with this data and a given set of hyperparameters, and its performance was evaluated on the full validation set.

This integrated approach allowed the data to determine the optimal trade-off between down-sampling the majority class and applying a cost-weight to the minority class.

2.3.3 The Optimized Sampling Strategy

The optimization process, run for 700 trials each for XGBoost and LightGBM, yielded a clear result. For the best-performing model, XGBoost, the winning strategy was to **use a sampling ratio of 1.0**, meaning that **100% of the non-bankrupt (healthy) firms were retained in the training set**.

This means the optimal strategy discovered by the search was to **use the entire training set of 496 660 observations with no down-sampling**. The final, winning model configuration therefore applied a ‘scale_pos_weight’ of **777.46** (i.e., 496 022/638) to the full training data. This finding suggests that, for this dataset and model architecture, providing the algorithm with all available information while heavily weighting the rare class is superior to any form of majority class down-sampling.

2.4 Hyperparameter Optimization and Evaluation Framework

A rigorous framework for hyperparameter optimization and evaluation was central to this thesis, ensuring that each model was tuned to its optimal potential and that the final methodological choices were supported by empirical evidence.

2.4.1 Hyperparameter Optimization Framework

All hyperparameter tuning was conducted using the **Optuna** framework, a state-of-the-art tool for automatic search space optimization. For all experiments, Optuna was configured to use its default sampler, the **Tree-structured Parzen Estimator (TPE)**. TPE is a sophisticated Bayesian optimization algorithm that effectively navigates the search space. It uses the performance of past trials to build a probabilistic model of the objective function, allowing it to focus

future trials on more promising hyperparameter regions and converge on a high-performing configuration more efficiently than a random or grid search. The number of trials varied by experiment, with up to **700 trials** per model being used for the final tuning in Phase 2.

2.4.2 Validation of the Optimization Metric

A key methodological choice for the extreme imbalance of Phase 2 was the selection of an appropriate optimization metric. While single-point metrics like the F-score are common, recent literature, such as Holzmann and Klar (2024), shows they are not robust to severe class imbalance. This led to a choice between the two primary curve-based metrics: ROC-AUC and PR-AUC. Contradicting the common belief that PR-AUC is superior for imbalanced data, recent work by McDermott et al. (2024) and Richardson et al. (2024) suggests ROC-AUC may be more stable in cases of extreme rarity.

To empirically validate the best objective for this specific dataset, a dedicated experiment was conducted. Four candidate metrics were tested: **ROC-AUC**, **PR-AUC**, **Partial PR-AUC** [0.90, 0.95], and **Precision@90Recall**. For each metric, a 100-trial Optuna optimization was performed on the XGBoost architecture. To ensure robustness, this entire process was repeated five times with different random seeds.

The performance of the resulting 20 models was assessed on the held-out validation and test sets, with the results summarized in Table 2.1 and Table 2.2. An Analysis of Variance (ANOVA) followed by a Tukey HSD post-hoc test was used to determine statistical significance.

Table 2.1: Performance Comparison - Valid Phase

Metric	Optimization Method			
	ROC-AUC	PR-AUC	Partial PR-AUC	Precision @90Recall
ROC-AUC	0.9479±0.0004***	0.9383±0.0035	0.9446±0.0004	0.9430±0.0029
PR-AUC	0.0186±0.0005	0.0249±0.0049***	0.0177±0.0010	0.0166±0.0016
Part. PR-AUC	0.0092±0.0002	0.0085±0.0005	0.0094±0.0002***	0.0094±0.0001
Prec@90Rec	0.0109±0.0002	0.0101±0.0007	0.0111±0.0002	0.0112±0.0002***

* p < 0.1, ** p < 0.05, *** p < 0.01 (ANOVA + Tukey HSD)

Bold values indicate significantly better performance than all other methods

Values: mean ± std (n=5)

The results provide a clear justification for our final methodological choice. While each optimization strategy successfully maximized its own target metric on the validation set (Table 2.1), a significant divergence in performance occurred on the unseen test set. To formally assess this, an **Analysis of Variance (ANOVA)** was first conducted to test the null hypothesis that the mean performance of the four optimization strategies was equal. This test

Table 2.2: Performance Comparison - Test Phase

Metric	Optimization Method			
	ROC-AUC	PR-AUC	Partial PR-AUC	Precision @90Recall
ROC-AUC	0.9447±0.0014**	0.9400±0.0024	0.9420±0.0014	0.9417±0.0025
PR-AUC	0.0147±0.0002**	0.0147±0.0011	0.0136±0.0004	0.0136±0.0012
Part. PR-AUC	0.0086±0.0005*	0.0082±0.0005	0.0071±0.0014	0.0069±0.0018
Prec@90Rec	0.0098±0.0003**	0.0090±0.0006	0.0096±0.0003	0.0096±0.0004

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (ANOVA + Tukey HSD)

Bold values indicate significantly better performance than all other methods

Values: mean $\pm$ std (n=5)

calculates the F-statistic, the ratio of the variance between the groups to the variance within them ($F = \frac{MSB}{MSW}$). If the resulting p-value was significant, a **Tukey HSD post-hoc test** was performed to compare every pair of optimization strategies, identifying which specific groups were significantly different from one another. As shown in Table 2.2, the models optimized using ROC-AUC not only were more stable but also showed statistically superior performance across all four evaluation metrics. This model achieved a test ROC-AUC of 0.9447 ± 0.0014 and a PR-AUC of 0.0147 ± 0.0002 , outperforming the models specifically tuned on PR-based metrics. This experiment empirically confirms that for this dataset, ROC-AUC is the most robust and effective optimization objective, and it was therefore selected for the final model tuning in this thesis.

Chapter 3

Empirical Results and Discussion

3.1 Phase 1: Performance on the Balanced Sample

3.1.1 Hyperparameter Optimization Insights

To ensure a fair and robust comparison, each of the six models underwent an extensive hyperparameter tuning process using the Optuna framework. The number of optimization trials was adjusted based on model complexity and computational time: 1000 trials each for XGBoost and LightGBM, 150 each for Random Forest, Neural Network, and SVM, and 100 for the simpler Logistic Regression. The objective for each model was to maximize the ROC-AUC score on the held-out validation set. The following analysis explores the key insights from this tuning process.

Random Forest

For the Random Forest model, the optimization process settled on an interesting configuration: a large ensemble of **1370** very shallow trees (`max_depth=7`) that considered all features at each split (`max_features=None`). As visualized in the optimization plot (Appendix, Figure A.1), this choice effectively turns the algorithm into a classic Bagging model, where the only source of randomness between trees is the bootstrap sampling of the data. This suggests that for this dataset, the variance reduction from averaging many lightly regularized trees built on high-signal data was sufficient, and additional randomness from feature subsampling was not beneficial.

Gradient Boosting Models (XGBoost & LightGBM)

The tuning of the two gradient boosting models revealed a well-defined optimal region characterized by a trade-off between a low `learning_rate` and a moderate number of estimators (`n_estimators`). The optimization paths (visualized in Appendix, Figures A.2 and A.3) show

that while the optimizer selected a single best set of parameters, multiple paths led to high-performing models. This indicates the presence of a stable valley in the loss landscape, where various combinations of low learning rates and simpler tree structures yield strong results.

Neural Network and Linear Models

The Neural Network tuning resulted in a relatively deep and narrow architecture of **7 layers** with **97 neurons** each. A notable finding from its optimization path (Appendix, Figure A.4) was the preference for a very large `batch_size` of **4096**, which can provide a regularizing effect and more stable gradient estimates.

For the linear and margin-based models, the key insight was the importance of strong regularization. The Logistic Regression model performed best with a very small `C` value of **0.0020**, indicating a high L1 penalty on coefficient magnitudes (Appendix, Figure A.6). Similarly, the SVM optimization overwhelmingly favored a simple **linear kernel** over the more complex RBF kernel, again with a small `C` value of **0.0158** (Appendix, Figure A.5). This shows that for this dataset, non-linear transformations offered no benefit and that controlling model complexity was paramount for these architectures.

3.1.2 Model Performance Comparison

Following hyperparameter tuning, the six optimized models were evaluated on the unseen balanced test set. Figure 3.1 shows the ROC curve comparison, which measures the overall discriminatory power of each model. All models perform well, with ROC-AUC scores above 0.91. However, the tree-based ensembles clearly stand out, with **RandomForest (AUC = 0.940)**, **LightGBM (AUC = 0.938)**, and **XGBoost (AUC = 0.936)** achieving the top scores.

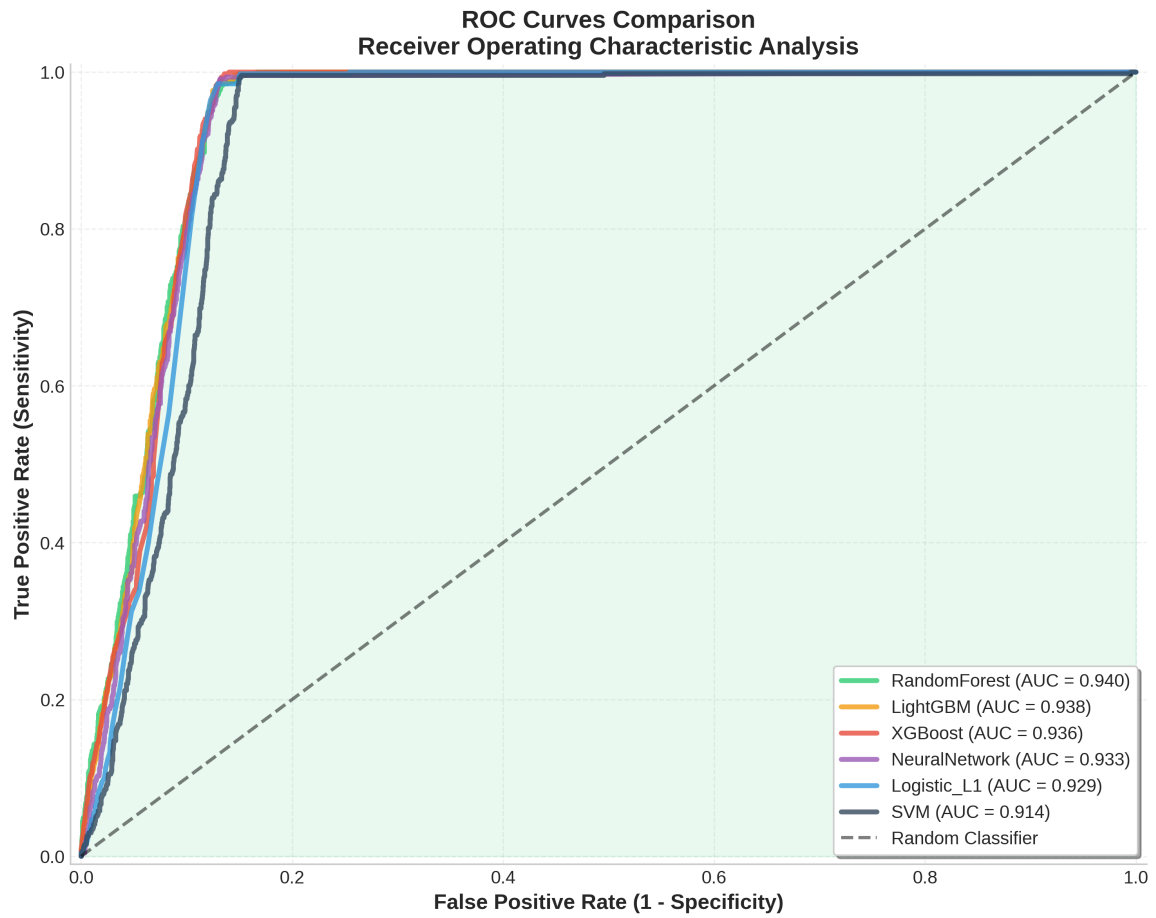


Figure 3.1: ROC curve comparison for the six optimized models on the balanced test set.

Further analysis of the model outputs provides deeper insights. The distribution of predicted probabilities, shown in Figure 3.2, reveals that the top-performing ensemble models and the Neural Network produce well-separated, bimodal predictions (scores cluster near 0.0 and 1.0). In contrast, the SVM appears far less confident, with its predictions clustered in the middle of the probability range.

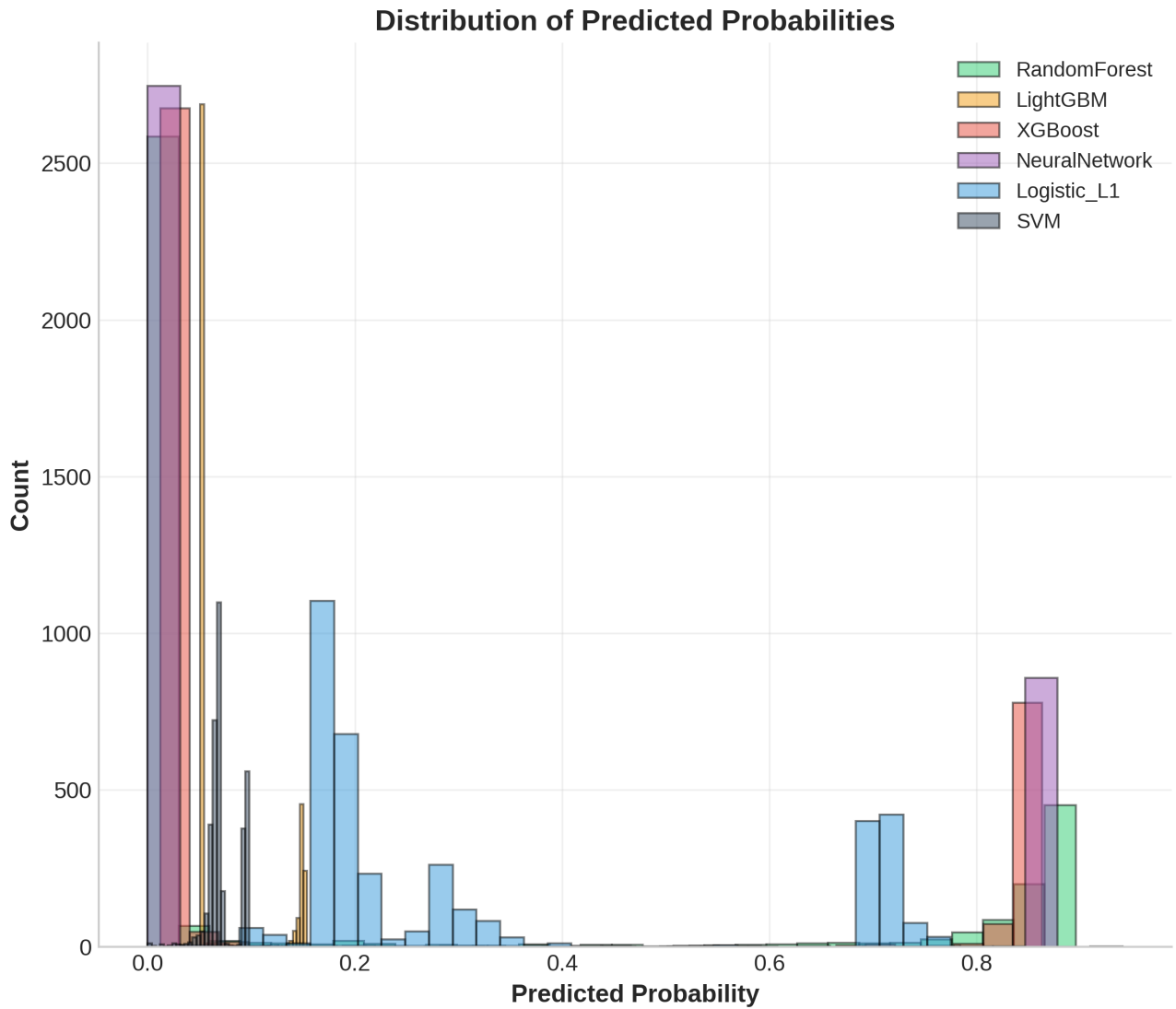


Figure 3.2: Distribution of predicted probabilities on the test set for each of the six models.

Finally, an analysis of the correlation between model predictions (Figure 3.3) shows that the top three ensemble models (RandomForest, LightGBM, XGBoost) and the Neural Network form a highly consistent "consensus bloc," with their probability scores having correlations above 0.98. This indicates they are learning very similar underlying patterns in the data. The SVM is again the outlier, showing the lowest correlation with the other models.

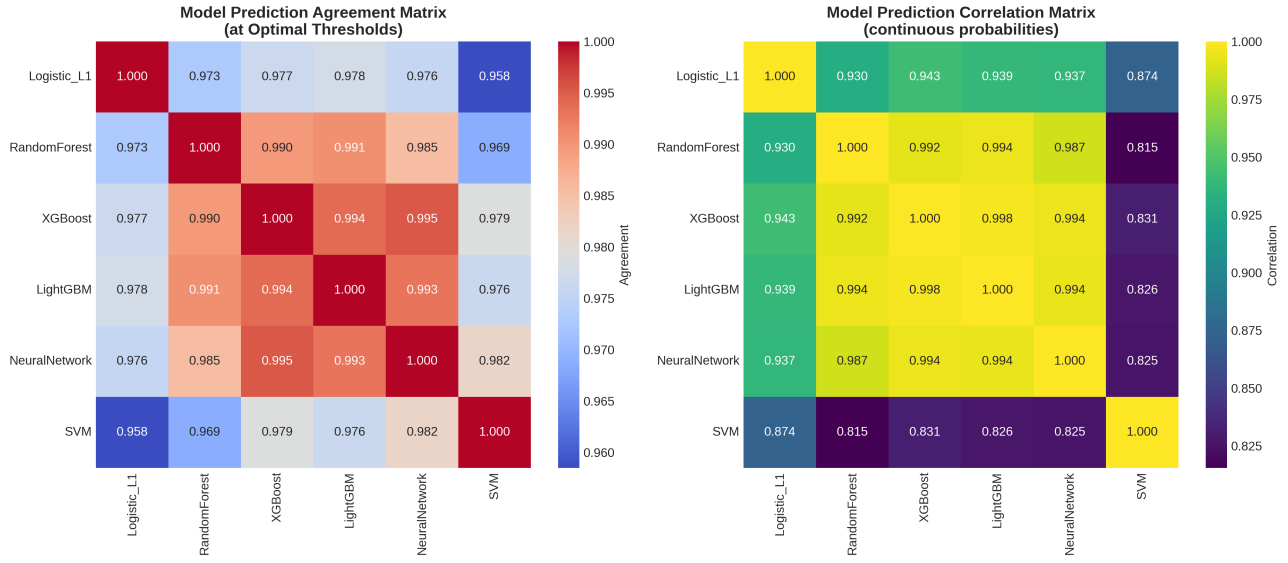


Figure 3.3: Model prediction agreement. Left: Agreement on binary classifications. Right: Pearson correlation of continuous probability scores.

Based on these results, the tree-based ensembles, particularly RandomForest, XGBoost, and LightGBM, were identified as the most effective models. Given the superior computational efficiency of XGBoost and LightGBM, they were selected as the final candidates for the more demanding full-population analysis in Phase 2.

3.2 Phase 2: Full Population Analysis

3.2.1 Hyperparameter Optimization of Final Models

The second phase of the analysis focused on tuning the two leading models, XGBoost and LightGBM, on the full, extremely imbalanced dataset. This involved an extensive optimization process of **700 trials** for each model using Optuna's TPE sampler. The optimization history plots (Figure 3.4 and 3.5) show the best ROC-AUC score found as the search progressed. For clarity, these plots show the top 95% of objective values, as the full history can be skewed by a few poorly performing outlier trials (the full history plots are available in Appendix A.2). The stability of the XGBoost model's performance over hundreds of subsequent optimization trials provides strong confidence that the search process successfully identified the optimal parameter configuration within the defined space.

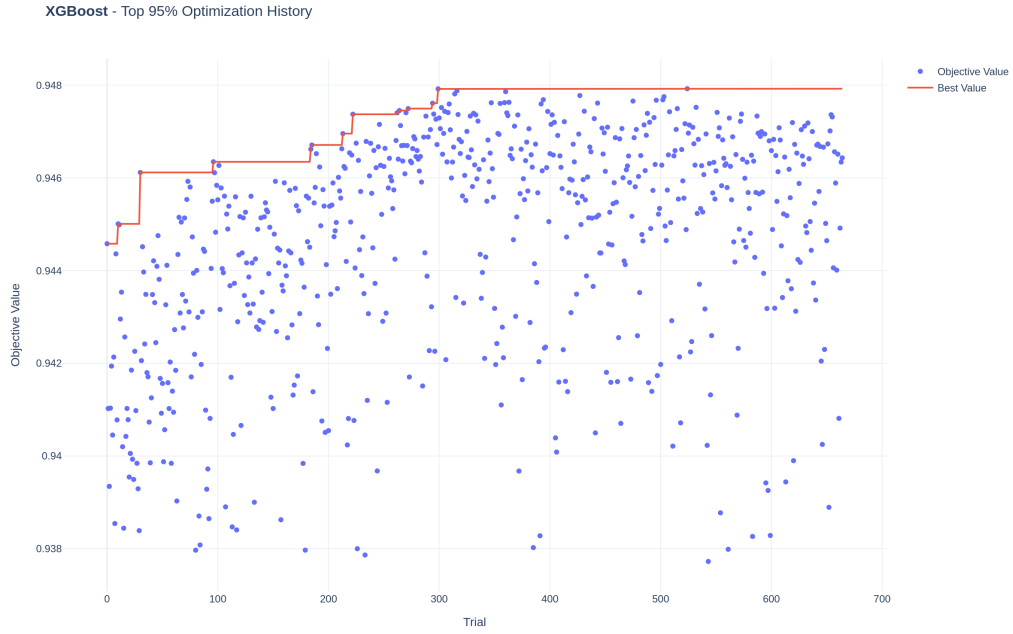


Figure 3.4: Optimization history for XGBoost, showing the best objective value (ROC-AUC) found as a function of the trial number. The plot is zoomed to the top 95% of trials for clarity.

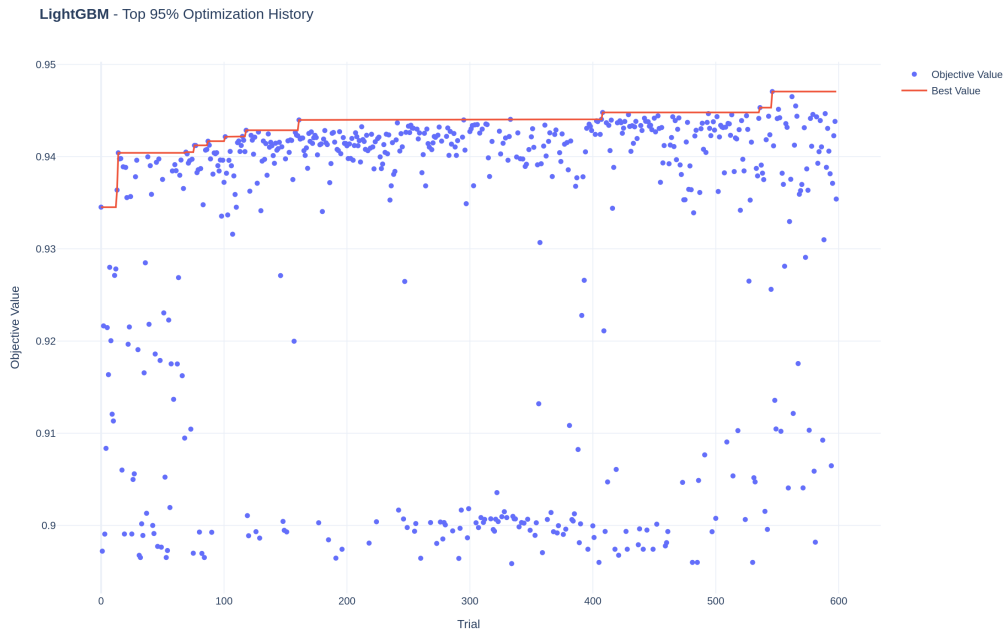


Figure 3.5: Optimization history for LightGBM (top 95% of trials).

A key part of this optimization was treating the majority class down-sampling ratio as a tunable hyperparameter. The parallel coordinate plots (Figures 3.6 and 3.7) visualize the entire search. For both models, the search space contained multiple regions of high performance.

For instance, the XGBoost plot shows that while the best trials converged on a specific set of parameters, other combinations involving different learning rates and tree depths also performed well. This highlights the complexity of the hyperparameter landscape. A critical finding is that for both models, the highest-performing trials consistently converged on high sampling ratios, with XGBoost showing a clear preference for a `sampling_ratio` of 1.0. The final parameters chosen for each model reflect the single best trial found during the search.

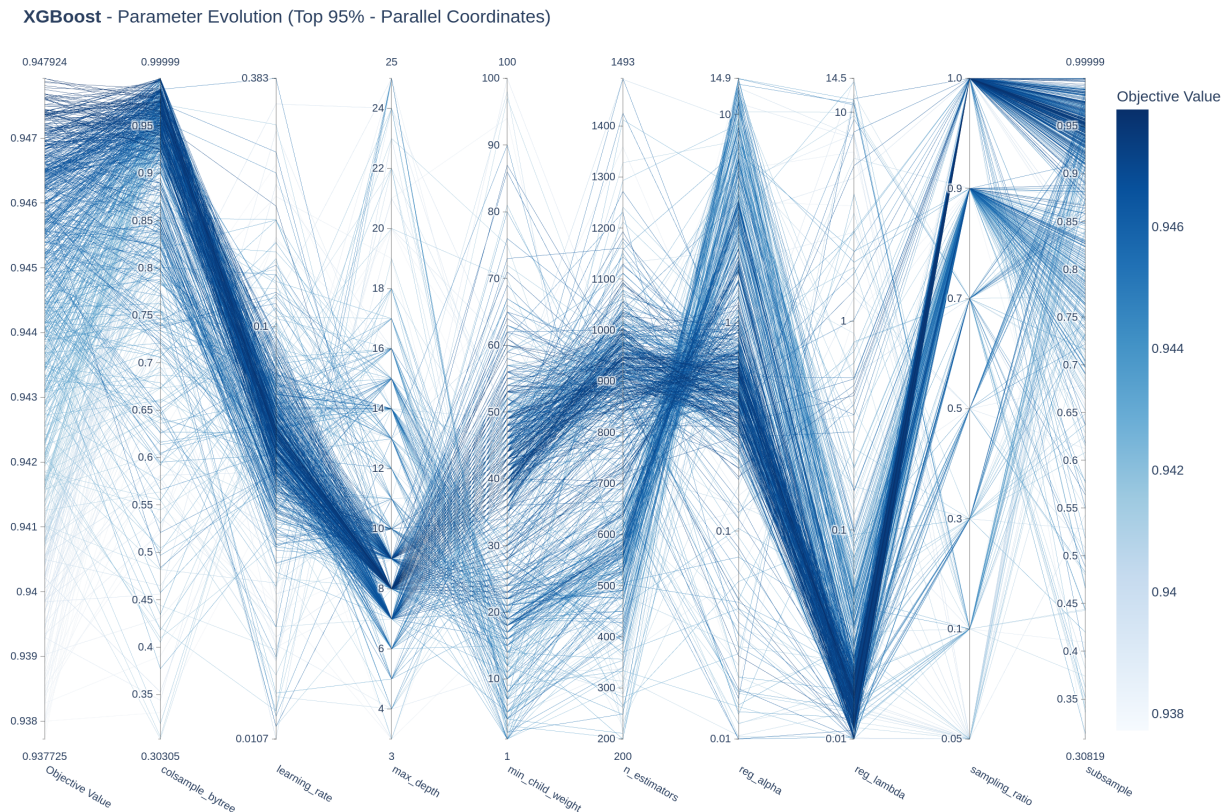


Figure 3.6: Parallel coordinate plot for the XGBoost optimization (top 95% of trials). The dark blue lines show a clear preference for a `sampling_ratio` of 1.0.

LightGBM - Parameter Evolution (Top 95% - Parallel Coordinates)

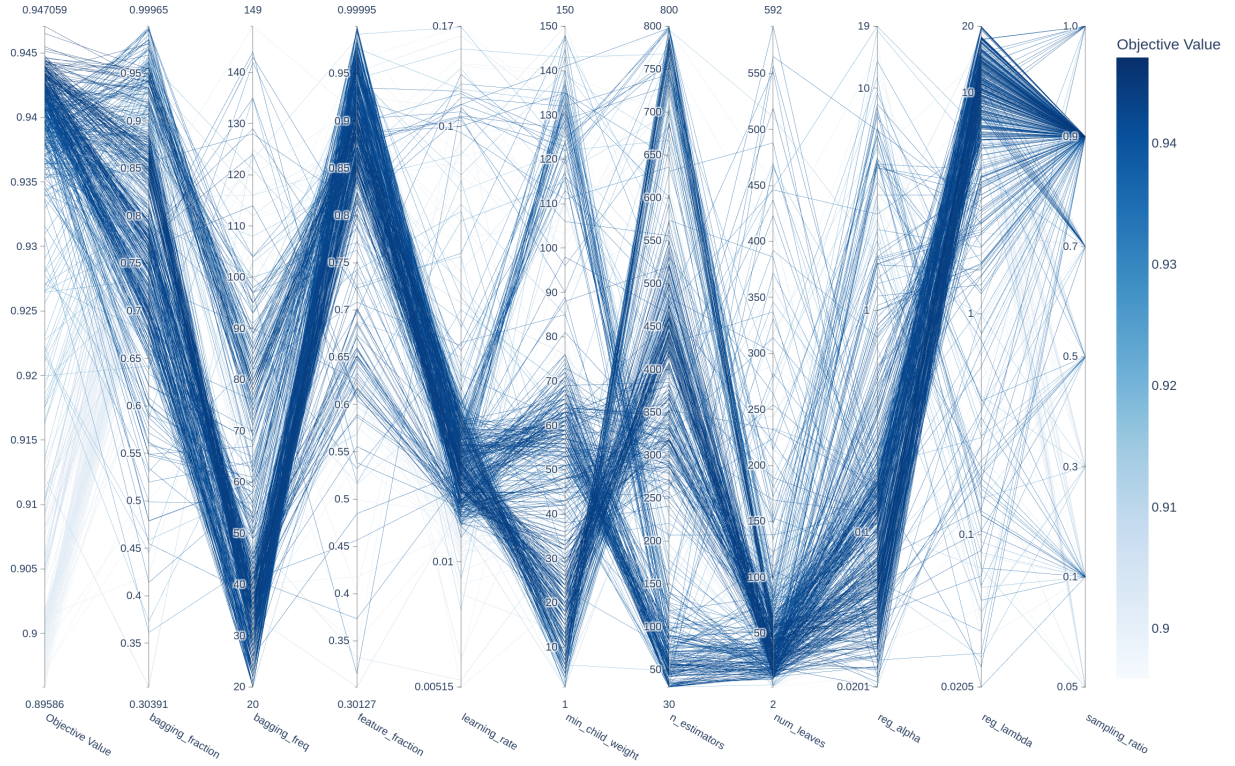


Figure 3.7: Parallel coordinate plot for the LightGBM optimization (top 95% of trials).

The final optimized hyperparameters for the two models were as follows: The final parameters chosen for each model reflect the single best trial found during the extensive optimization search.

XGBoost:

- colsample_bytree: 0.9647
- learning_rate: 0.0877
- max_depth: 8
- min_child_weight: 47
- n_estimators: 908
- reg_alpha: 0.3225
- reg_lambda: 0.0100
- subsample: 0.9537

LightGBM:

- bagging_fraction: 0.9405
- bagging_freq: 78
- feature_fraction: 0.9061
- learning_rate: 0.0187
- min_child_weight: 4
- n_estimators: 442
- num_leaves: 34
- reg_alpha: 0.0776
- reg_lambda: 11.6802

3.2.2 Final Model Performance

After separate tuning, the final optimized XGBoost and LightGBM models were compared head-to-head on the unseen test set. The ROC curves in Figure 3.8 show that both models

are highly performant, but **XGBoost** achieves a slightly superior ROC-AUC of 0.947, compared to 0.942 for LightGBM. Based on this result, XGBoost was selected as the final, champion model for all subsequent analysis.

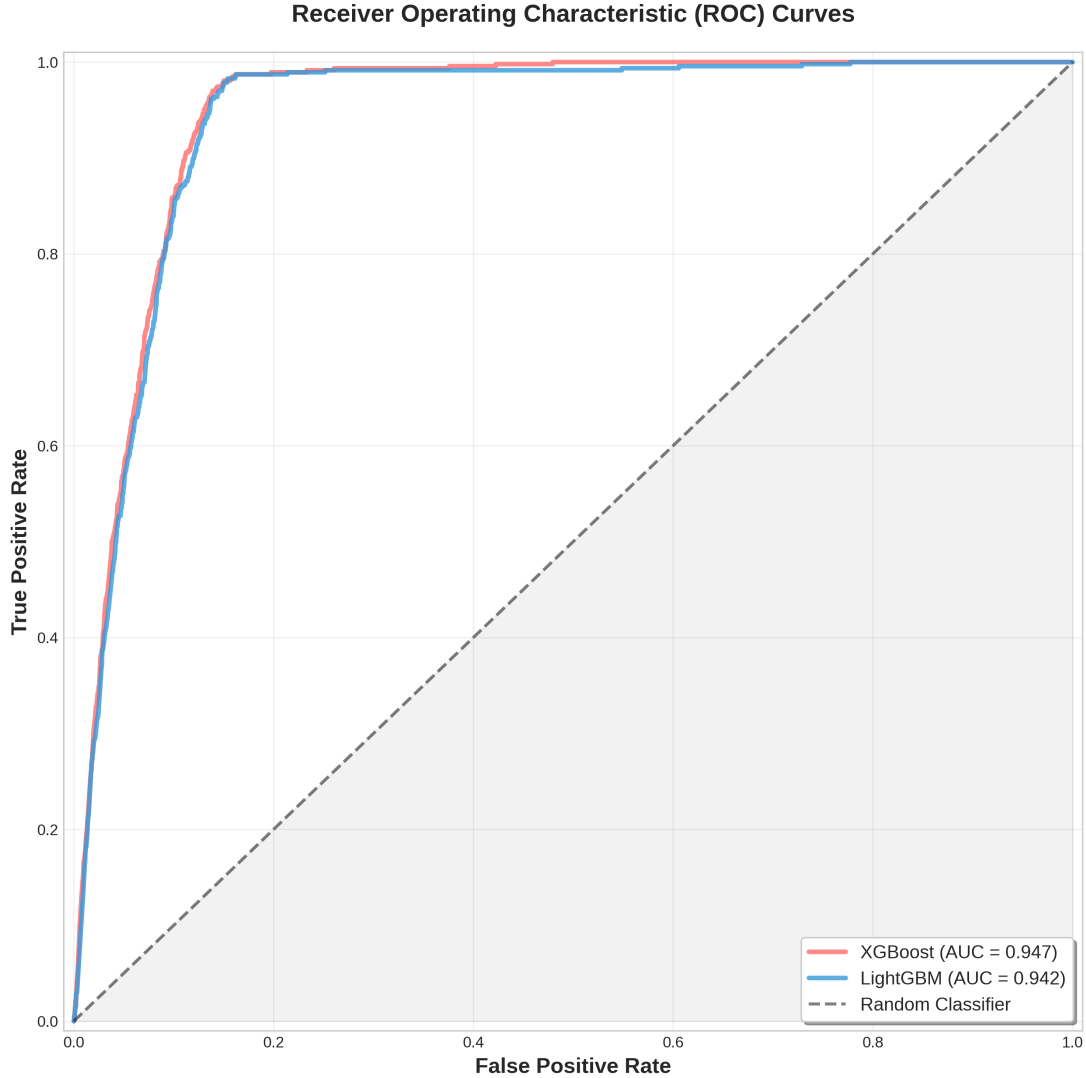


Figure 3.8: Final ROC curve comparison on the full test set. The tuned XGBoost model (AUC = 0.947) slightly outperforms the LightGBM model (AUC = 0.942).

To understand how the final XGBoost model works, Figure 3.9 shows the distribution of its predicted bankruptcy probabilities. The plot reveals an excellent separation: the vast majority of healthy firms receive a score very close to zero (mean = 0.141), while bankrupt firms receive a much higher score (mean = 0.863). While this density plot effectively shows the model's ability to separate the two classes, its y-axis is normalized independently for each class. This can visually mask the scale of the classification challenge; the difficulties in managing false positives become more apparent when viewing the raw score distributions, as shown in the next section.

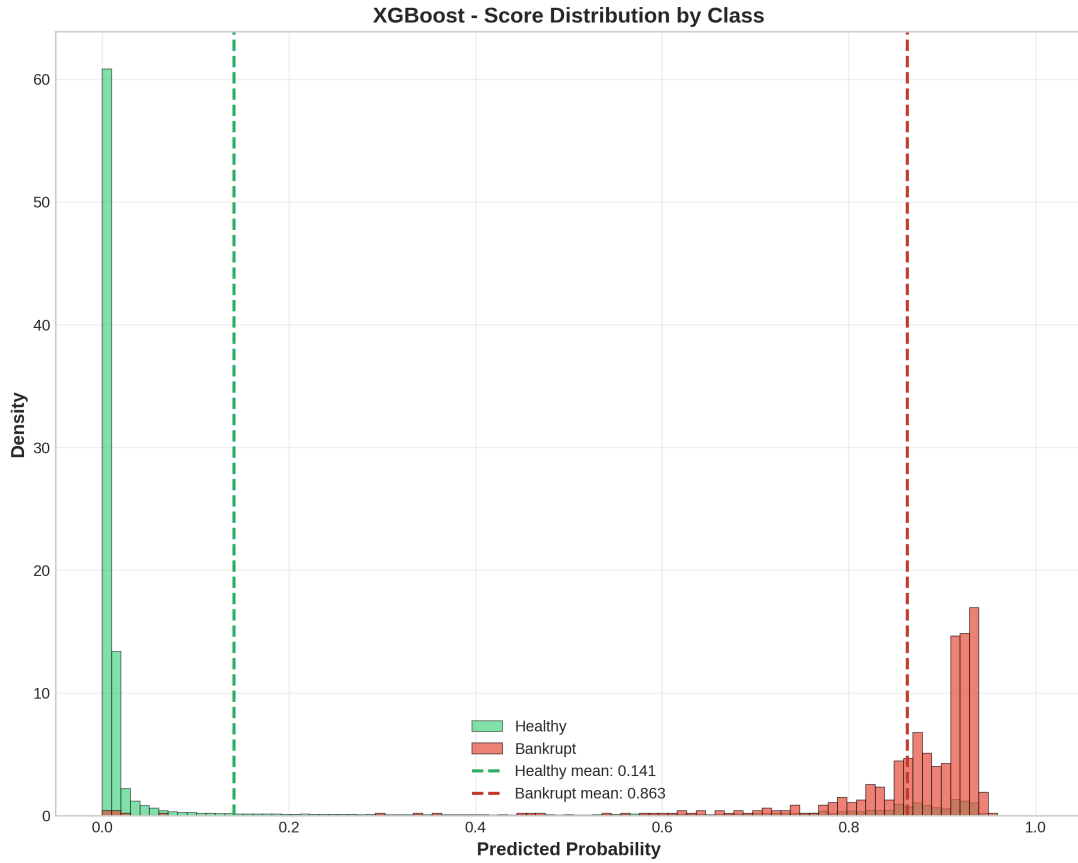


Figure 3.9: Score distribution by class for the final XGBoost model on the test set, showing clear separation between healthy and bankrupt firms.

3.2.3 Analysis of Practical Operating Points

While the histogram shows good separation, a boxplot of the scores (Figure 3.10) better illustrates the practical challenge. To capture a very high percentage of bankruptcies (i.e., achieve high recall), one must lower the decision threshold, which inevitably flags more healthy firms as high-risk (false positives). The plot shows that the thresholds required to achieve 90% and 95% recall are in a reasonable range, but achieving 99% recall requires a very low threshold that leads to a significant overlap with the healthy population.

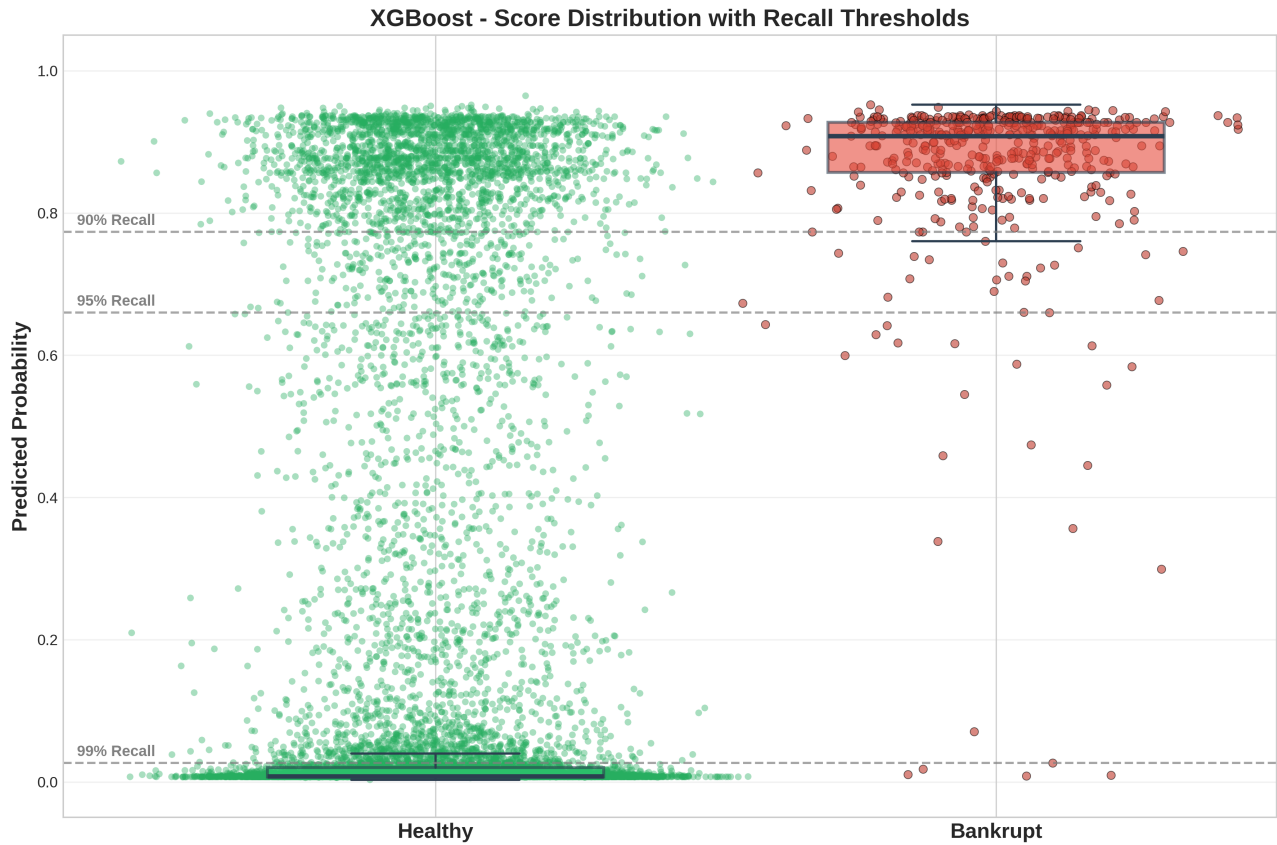


Figure 3.10: Boxplot of predicted scores showing the thresholds required to achieve 90%, 95%, and 99% recall on the test set.

This trade-off is quantified in the confusion matrices shown in Figure 3.11. At a recall target of **90.1%**, the model correctly identifies 421 of the 467 actual bankruptcies. To achieve this, it flags 40,889 healthy firms, meaning a user would need to investigate just **11.1%** of the healthy population. To increase recall to **95.1%**, the number of healthy firms to investigate rises to 13.0%. Given that a random selection would require investigating 90% of firms to find 90% of bankruptcies, the model provides a roughly **eight-fold efficiency gain**.

Furthermore, the model provides significant value by creating a large pool of firms predicted to be healthy with a much lower risk profile. At the 90% recall threshold, the bankruptcy rate among the 326,184 firms predicted as "Healthy" is just **0.014%** (46 failures). This is a nine-fold reduction from the original population's base rate of 0.127%, demonstrating the model's ability to effectively screen out low-risk companies.

XGBoost - Confusion Matrices at 90% & 95% Recall Targets

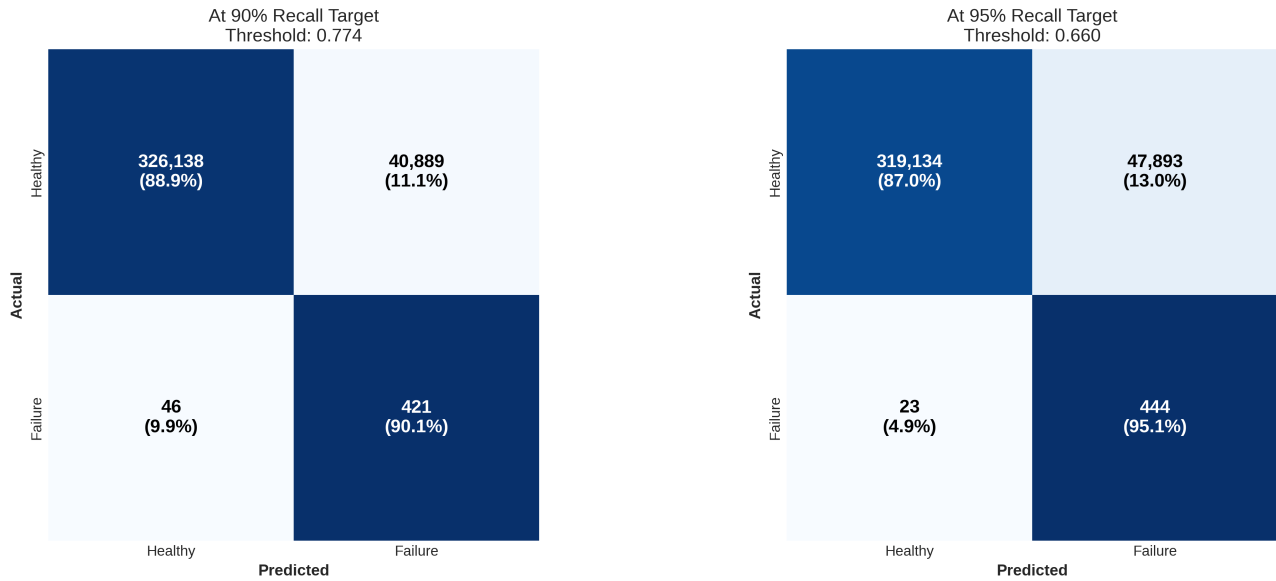


Figure 3.11: Confusion matrices for the final XGBoost model at 90% and 95% recall targets on the test set.

3.2.4 Model Feature Importance

While the primary goal of this thesis is predictive accuracy, a benefit of using a tree-based model like XGBoost over a "black box" like a neural network is its interpretability. By examining which features the model relied on most heavily, it is possible to gain insights into the drivers of bankruptcy risk.

Figure 3.12 shows importance by "split count," or how frequently a feature is used to make a decision in the model's trees. The most frequently used features are related to the company's age (`Year_from_incorporation`) and, critically, its geographic location (`Registered_office_address_Latitude` and `Longitude`).

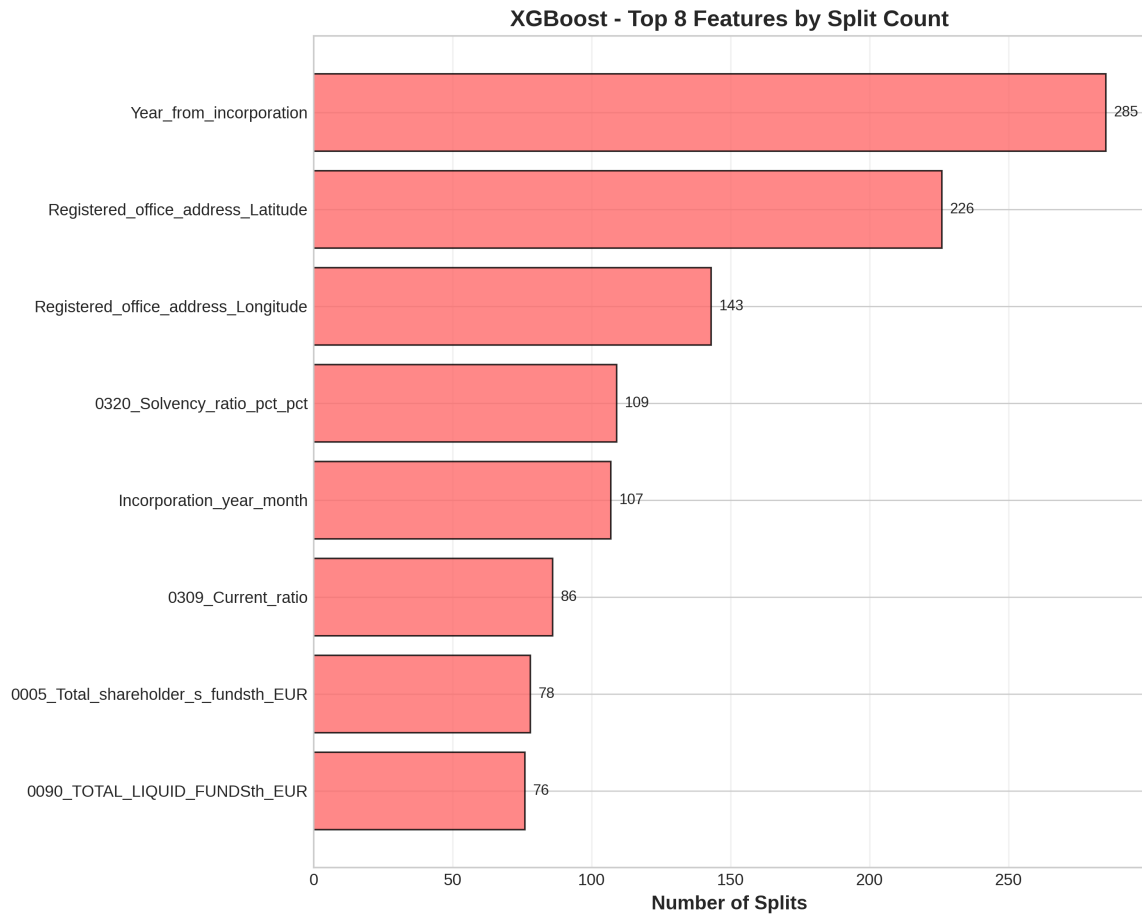


Figure 3.12: Top 8 features by number of splits for the final XGBoost model.

The high importance of these geographic features suggests that local economic patterns are strongly correlated with bankruptcy risk. Figure 3.13 visualizes this finding by plotting the decision boundaries learned by the model directly onto a map of Italy. The red lines indicate the latitude and longitude values where the XGBoost model made splits, with the thickness of the lines corresponding to the importance (gain) of that split. The map clearly shows that the model learned to partition Italy into distinct risk regions, confirming that geography is a key, non-linear driver of bankruptcy prediction in this dataset.

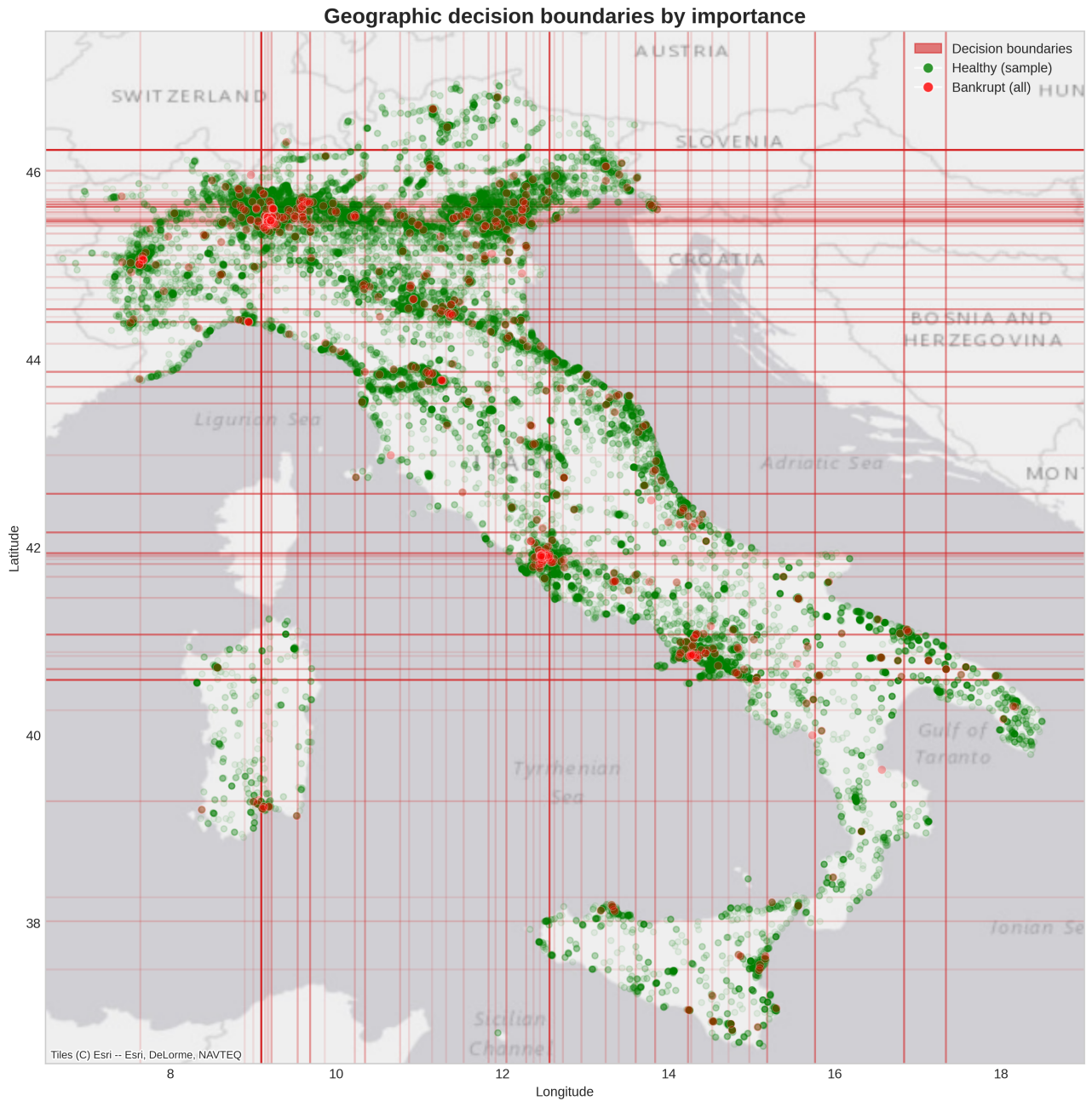


Figure 3.13: Geographic decision boundaries learned by the XGBoost model. The red lines show where the model splits the data by latitude and longitude, with line thickness proportional to the split's importance (gain).

Chapter 4

Conclusion

4.1 Summary of Key Findings

This thesis developed and assessed a machine learning framework for one-year-ahead corporate bankruptcy prediction in Italy, confronting the core challenge of extreme class imbalance where failures represent just 0.127% of firm-year observations. The research was structured in two distinct phases to systematically address this problem.

Phase 1 established a performance baseline by comparing six different machine learning algorithms on a balanced pilot sample. The results indicated the superiority of tree-based ensemble methods, with Random Forest, XGBoost, and LightGBM significantly outperforming linear models and neural networks in terms of discriminatory power.

Phase 2 extended the two most efficient models, XGBoost and LightGBM, to the full, unfiltered population of approximately 190,000 enterprises. This phase included two key methodological components. First, the optimization strategy treated the down-sampling ratio of the majority class as a hyperparameter to be tuned alongside model parameters. The search process concluded that the optimal approach was not to down-sample at all, but to use 100% of the healthy firm data while applying a large minority class weight (`scale_pos_weight = 777.46`). Second, a dedicated experiment was conducted to validate the choice of an optimization metric. This experiment showed that optimizing for ROC-AUC produced more stable and higher-performing models on the unseen test data compared to metrics like PR-AUC.

The final, tuned XGBoost model emerged as the best model, achieving a test ROC-AUC of **0.947**. From a practical standpoint, this model offers an eight-fold efficiency gain over random selection: to identify 90% of all bankruptcies, an institution would only need to investigate the 11.1% of firms flagged as highest risk. Furthermore, the model effectively screens out low-risk companies, reducing the bankruptcy rate in the population of predicted-healthy firms to just 0.014%. Finally, an analysis of the model's feature importances revealed that while financial variables were key drivers of performance, geographic coordinates were among the

most frequently used features for decision splits, suggesting that local economic patterns are a significant, non-linear predictor of bankruptcy risk in Italy.

4.2 Limitations and Future Research Directions

While this thesis successfully develops a robust framework for bankruptcy prediction, it is important to acknowledge certain limitations that also present clear avenues for future research. The current study employs a fixed temporal split for training and testing. A valuable extension would be to implement a more dynamic evaluation, such as a **rolling-window or expanding-window validation**. This would provide deeper insights into the model’s temporal stability, its performance across different economic cycles, and the optimal frequency for retraining in a real-world production environment.

Furthermore, this study intentionally focused on the traditional **Fallimento** definition of bankruptcy to ensure a consistent target variable. The most critical next step is to adapt and extend the model to predict the newer **Liquidazione Giudiziale (Judicial Liquidation)** outcome. This would involve carefully analyzing the data collected since the recent Italian legal reforms and would make the model directly relevant to the current insolvency landscape.

Finally, while the metric selection experiment provided strong evidence for the use of ROC-AUC in this context, the broader academic debate is ongoing. Further research could explore the performance of these optimization metrics on different datasets or with other model architectures, contributing to a more comprehensive understanding of how to best evaluate classifiers in settings of extreme class imbalance.

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